

Fanyun Fang

Case Western Reserve University, Cleveland, OH, 44106, USA

Email: fxf116@case.edu Phone: +1(216)312-3747

EDUCATION

Case Western Reserve University	Present
Graduate Non-degree	
GPA: 4.0/4.0	
Case Western Reserve University	December 2025
M.S. in Medical Physiology	
GPA: 3.52/4.0	
Kent State University	August 2022
B.S. in Molecular/Cellular Biology, Minor in Anthropology	
GPA: 3.3/4.0	

EXPERIENCE

Research Technician III	09/2025 – Present
Department of Pharmacology, Case Western Reserve University	37.5 Hour / Week
Advisor: Dr. Chinyere Iweka	

QPRT Project

- Studied the role of QPRT in regulating the kynurenine-NAD⁺ pathway during aging and post-stroke immune dysfunction using myeloid-specific knockout and overexpression mouse models.
- Performed metabolic and mitochondrial phenotyping (Seahorse assays, Western blot) and immune profiling via flow cytometry and cytokine analysis.
- Linked cellular metabolic alterations to functional outcomes through microglial models and behavioral assays (Open field, NOR, MWM) of cognition and motor function.

HSC Project

- Investigated how BMAL1-dependent circadian regulation influences long-term hematopoietic stem cell (LT-HSC) abundance, lineage commitment, and functional reconstitution using WT and HSC-specific Bmal1 knockout mice.
- Optimized LT-HSC isolation and flow cytometric sorting and perform colony-forming unit assays to evaluate circadian phase- and genotype-dependent differences in hematopoietic lineage potential.
- Integrated time-resolved ATAC-seq, RNA-seq, CUT&RUN, and competitive transplantation to examine chromatin accessibility, gene expression, transcription-factor occupancy, and long-term donor chimerism.

Graduate Research Intern

02/2025 – 06/2025

Genomic Medicine Institute, Cleveland Clinic

20 Hour / Week

Advisor: Dr. Feixiong Cheng and Dr. Zhigang Liu

- Cultured and maintained SH-SY5Y neuroblastoma cell line and differentiated SH-SY5Y neuroblastoma cell line to neuron cells. Evaluated toxicity of fluoxetine, gabapentin, and budesonide in differentiated neuron cells using Bio-Rad protein quantification assays.
- Studied the effectiveness of these drugs in eliminating phosphorylated tau (p-tau) in differentiated neuron cells by using Western blot analysis to evaluate protein expression level of total tau and p-tau.
- Cultured and maintained induced pluripotent stem cell (iPSC) lines derived from healthy donors under sterile culture conditions.

Undergraduate Research Volunteer

09/2021 – 05/2022

Biology Department, Kent State University

20 Hour / Week

Advisor: Aleisha Moore

- Used chemo genetics model to determine the mechanism by which KNDy (kisspeptin/neurokinin B/dynorphin) cells and associated cell populations generate and regulate GnRH (gonadotrophin-releasing hormone) pulses.
- Manipulated microtome sectioning and immunostaining to observed differentiation in wild type mice brains and with changes in KNDy neurons mice brains.

Undergraduate Individual Investigation

Biology Department, Kent State University

Advisor: Dr. Jennifer McDonough

Taken with Individual Investigation Credits

01/2022 – 05/2022

20 Hour / Week

- Measured changes in mitochondria and in myelin in mice brain and spinal cord tissue from mice models of multiple sclerosis.
- Performed immunohistochemistry and Western blotting to measure tissue damage and the effects of therapies on repairing this damage.
- Observed under confocal and compared oligodendrocyte differentiation and remyelination with different treatments. (Cell Cultures manipulated under sterile environment).

Product Support Intern

ABcolonial Technology

09/2021 – 12/2021

20 Hour / Week

- Priced company's products and conducted market research on products used in biotechnology to match the positioning of the product in the market.
- Assisted the sales department with information about experiments and applications of the product to satisfy the customer's demands and the need to buy the appropriate product.
- Wrote biological blogs to the company to provide knowledge about diseases and effects of medicine to the public.

SKILLS

Experimental: PCR, Immunostaining, Western Blot, Microtome sectioning, Cell Cultures, Confocal

Software: InDesign, Photoshop, Premiere